A.

A-1.

$$A = \begin{pmatrix} 4 & 0 \\ 3 & -5 \end{pmatrix}$$

① A^TA 계산

$$\begin{pmatrix} 4 & 3 \\ 0 & -5 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 3 & -5 \end{pmatrix} = \begin{pmatrix} 25 & -15 \\ -15 & 25 \end{pmatrix}$$

② λ 계산

$$\det(A^TA - \lambda I) = (25-\lambda)^2 - 225 = 0$$

$$\lambda^2 - 50\lambda + 400 = 0$$

$$\lambda = 40, 10$$

③ 특이값

$$\sigma_1 = \sqrt{40}, \quad \sigma_2 = \sqrt{10}$$

④ V

1. $\lambda = 40$

$$\begin{pmatrix} -15 & -15 \\ -15 & -15 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$x_1 = -x_2$$

$$s\begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

2. $\lambda = 10$

$$\begin{pmatrix} 15 & -15 \\ -15 & 15 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$x_1 = x_2$$

$$s\begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$\therefore A^TA = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}\begin{pmatrix} 40 & 0 \\ 0 & 10 \end{pmatrix}\begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

$$= P \cdot D \cdot P^T$$

$$(V = P)$$

⑤ U

$$u_1 = \frac{1}{\sigma_1} A v_1$$

$$= \frac{1}{\sqrt{40}} \cdot \begin{pmatrix} 4 & 0 \\ 3 & -5 \end{pmatrix}\begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

$$= \frac{1}{\sqrt{40}} \cdot \begin{pmatrix} 2\sqrt{2} \\ 4\sqrt{2} \end{pmatrix}$$

$$u_2 = \frac{1}{\sigma_2} A v_2$$

$$= \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} 4 & 0 \\ 3 & -5 \end{pmatrix}\begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$= \frac{1}{\sqrt{10}} \begin{pmatrix} 2\sqrt{2} \\ -\sqrt{2} \end{pmatrix}$$

$$\therefore U = \begin{pmatrix} \frac{2}{\sqrt{20}} & \frac{2}{\sqrt{5}} \\ \frac{4}{\sqrt{20}} & -\frac{1}{\sqrt{5}} \end{pmatrix}$$

$$\therefore A = U \Sigma V^T$$

$$= \begin{pmatrix} 2/\sqrt{20} & 2/\sqrt{5} \\ 4/\sqrt{20} & -1/\sqrt{5} \end{pmatrix}\begin{pmatrix} \sqrt{40} & 0 \\ 0 & \sqrt{10} \end{pmatrix}\begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

A-2.

$$A = \sigma_1 u_1 v_1^T \qquad 2\times1 \quad 1\times2$$

$$= \sqrt{40} \cdot \begin{pmatrix} 2/\sqrt{20} \\ 4/\sqrt{20} \end{pmatrix}\begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

$$= \sqrt{40}\begin{pmatrix} \frac{2}{\sqrt{40}} & -\frac{2}{\sqrt{40}} \\ \frac{4}{\sqrt{40}} & -\frac{4}{\sqrt{40}} \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -2 \\ 4 & -4 \end{pmatrix}$$

B

B-1.

$L(\lambda | (2,3,5,6))$

$= L(\lambda|2) \cdot L(\lambda|3) \cdot L(\lambda|5) \cdot L(\lambda|6)$

$= P(2|\lambda) \cdot P(3|\lambda) \cdot P(5|\lambda) \cdot P(6|\lambda)$

$= \lambda^2 \cdot 2 \cdot e^{-2\lambda} \cdot \lambda^2 \cdot 3 e^{-3\lambda} \cdot \lambda^2 \cdot 5 \cdot e^{-5\lambda} \cdot \lambda^2 \cdot 6 \cdot e^{-6\lambda}$

$= \lambda^8 \cdot 180 \cdot e^{-16\lambda}$

$\log L(\lambda)$

$= 180 \log \lambda^8 \cdot e^{-16\lambda}$

$= 180 (8 \cdot \log \lambda - 16\lambda)$

$\frac{d}{d\lambda} \log L = 180 \left(8 \cdot \frac{1}{\lambda} - 16 \right)$

$\therefore \lambda = \frac{1}{2}$ //

C.

C-1.

(1) $f(x,y) = x^2 y + \sin y$

$x = r\cos\theta, \quad y = r\sin\theta$

$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r}$

$\frac{\partial f}{\partial x} = 2xy \qquad \frac{\partial x}{\partial r} = \cos\theta$

$\frac{\partial f}{\partial y} = x^2 + \cos y \qquad \frac{\partial y}{\partial r} = \sin\theta$

$\therefore \frac{\partial f}{\partial r} = 2xy \cdot \cos\theta + (x^2 + \cos y) \sin\theta$

$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \theta}$

$\frac{\partial x}{\partial \theta} = -r\sin\theta, \quad \frac{\partial y}{\partial \theta} = r\cos\theta$

$\therefore \frac{\partial f}{\partial \theta} = 2xy \cdot (-r\sin\theta) + (x^2 \cos y) \cdot r\cos\theta$

(2) $r=1, \ \theta = \frac{\pi}{2}$

$x = 0, \quad y = 1$

$\therefore \frac{\partial f}{\partial r} = \cos 1$

$\frac{\partial f}{\partial \theta} = 0$

C-2.

(1)

$f(x,y,z) = xy^2 - 4x^2 y + z^2$

$\frac{\partial f}{\partial x} = y^2 - 8xy \,|_{(1,-1,2)} = 9$

$\frac{\partial f}{\partial y} = 2xy - 4x^2 \,|_{(1,-1,2)} = -6$

$\frac{\partial f}{\partial z} = 2z \,|_{(1,-1,2)} = 4$

$\frac{1}{\eta} \cdot w = \left(\frac{6}{\eta}, \frac{2}{\eta}, \frac{3}{\eta} \right)$

$\therefore (9, -6, 4) \cdot \left(\frac{6}{\eta}, \frac{2}{\eta}, \frac{3}{\eta} \right)$

$= \frac{54 - 12 + 12}{\eta} = \frac{54}{\eta}$

(2)

$\nabla f = (9, -6, 4)$

최대 방향도함수: $\|\nabla f\|$

$\|\nabla f\| = \sqrt{81+36+16}$

$= \sqrt{133}$

방향 벡터: $\frac{1}{\sqrt{133}}(9, -6, 4)$

C-3.

(1) 변수 x: A의 수

y: B의 수

목적 함수: $f = 3x + 4y$

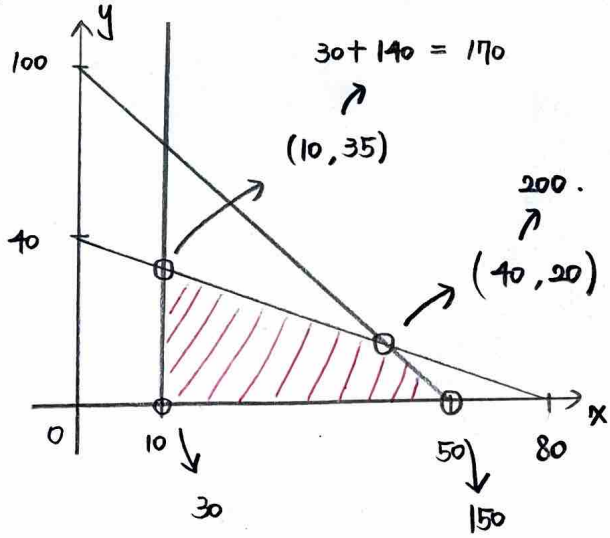
제약 조건: $2x+y \leq 100$,

$x+2y \leq 80$,

$x \geq 10$,

$x, y \geq 0$

(2)



A: 40개, B: 20개 일 때

200만원으로 이익 최대화.